

A
Project Report
On
NIRMAN AI

Submitted to

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For

The Subject

Applied AI and Prototype Development

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1. Introduction

NIRMAN AI is an AI-based road monitoring and decision-support system designed to identify road damage and track how road condition changes over time. The current prototype uses road photographs uploaded by a user or field engineer. Computer Vision detects visible defects such as cracks and potholes, while relevant road information can be used to estimate a road health score.

The main additional feature is deterioration tracking. The system stores inspection results for the same 50-meter road segment and compares them during later inspections. This helps identify whether the segment is deteriorating slowly, moderately, or rapidly.

The current prototype is photo-based so the AI pipeline can be developed and tested with open datasets and sample inspections. In future, cameras mounted on government inspection vehicles can collect road images automatically while the vehicle travels along the road.

1.1 Problem Statement

- Road inspection is largely manual and time-consuming.
- Cracks and potholes can increase between inspection cycles.
- A current road-condition score alone does not show how fast a road is deteriorating.
- NIRMAN AI detects road damage and calculates a health score for a 50-meter segment.
- Repeated inspection results are compared to calculate the deterioration rate.
- The system generates a Road Health Card to support maintenance prioritization.

1.2 Objective

- Detect visible cracks and potholes using Computer Vision.
- Estimate road condition using damage and relevant road information.
- Track the same 50-meter road segment across repeated inspections.
- Calculate deterioration rate and trend over time.
- Generate a Road Health Card for inspection and maintenance support.
- Develop the prototype using open datasets and free resources.

1.3 Technology Stack

Component	Technology	Purpose
Frontend	React.js / HTML / CSS	Image upload and interface
Backend	Python, FastAPI	Application logic and APIs
Computer Vision	YOLO, OpenCV, PyTorch	Crack and pothole detection
Machine Learning	Scikit-learn / XGBoost	Health and risk estimation
Database	PostgreSQL	Inspection and history storage
Training	Google Colab / Kaggle	Model training

2. Project Description

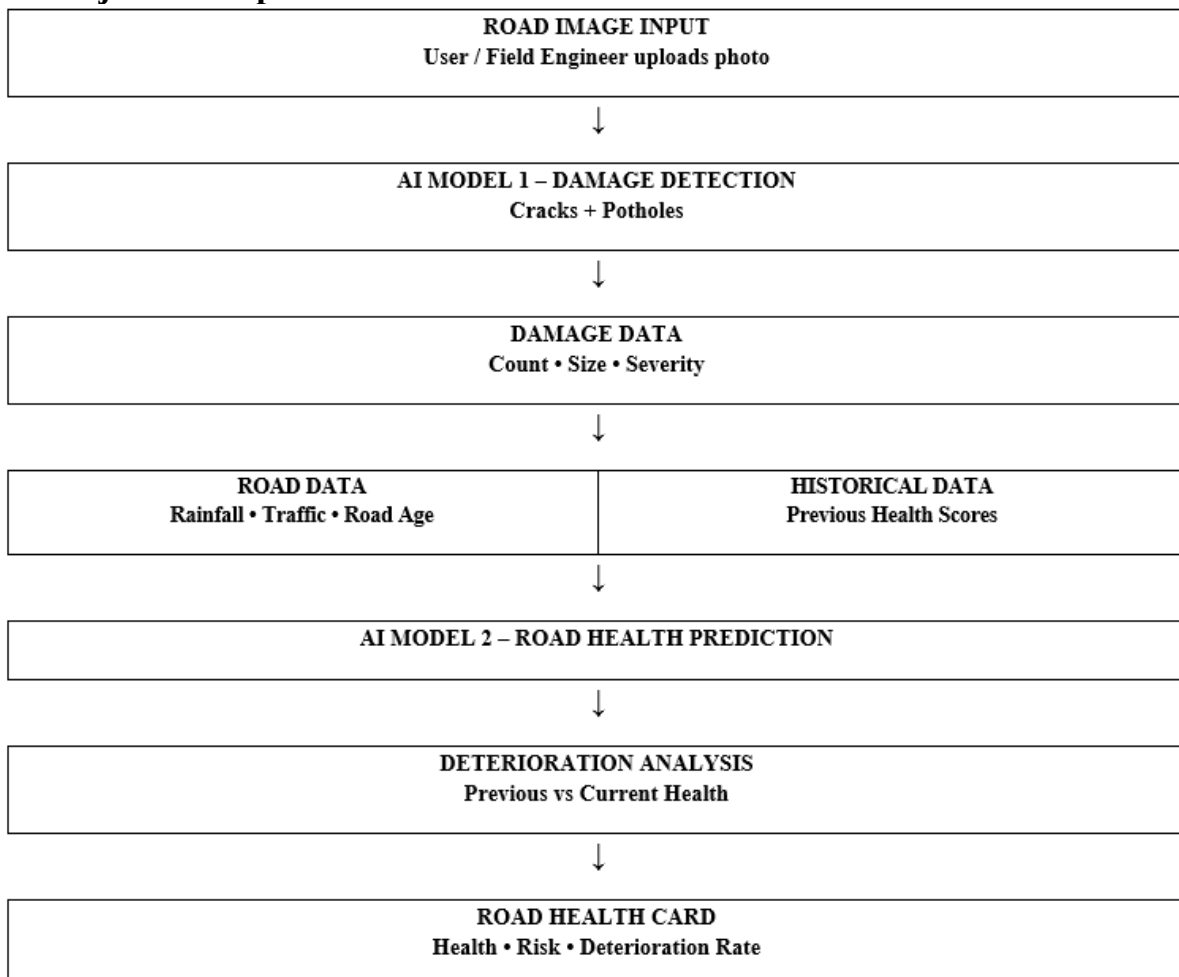
The project works in two main AI stages. First, a Computer Vision model analyzes an uploaded road image and detects visible damage such as cracks and potholes. The system extracts damage information such as count, approximate size and severity.

Second, damage information can be combined with rainfall, traffic, road age and maintenance information to estimate current road condition. The result is stored for the relevant 50-meter road segment.

During a later inspection, the new result is compared with the previous result. For example, if health changes from 82 to 68 and then to 49, the system identifies a fast deterioration trend. Therefore, NIRMAN AI does not only ask how healthy the road is today; it also tracks how quickly the same segment is getting worse.

The final output is a Road Health Card containing current health, detected damage, risk level and deterioration trend. The prototype is a decision-support system and does not replace qualified engineering inspection.

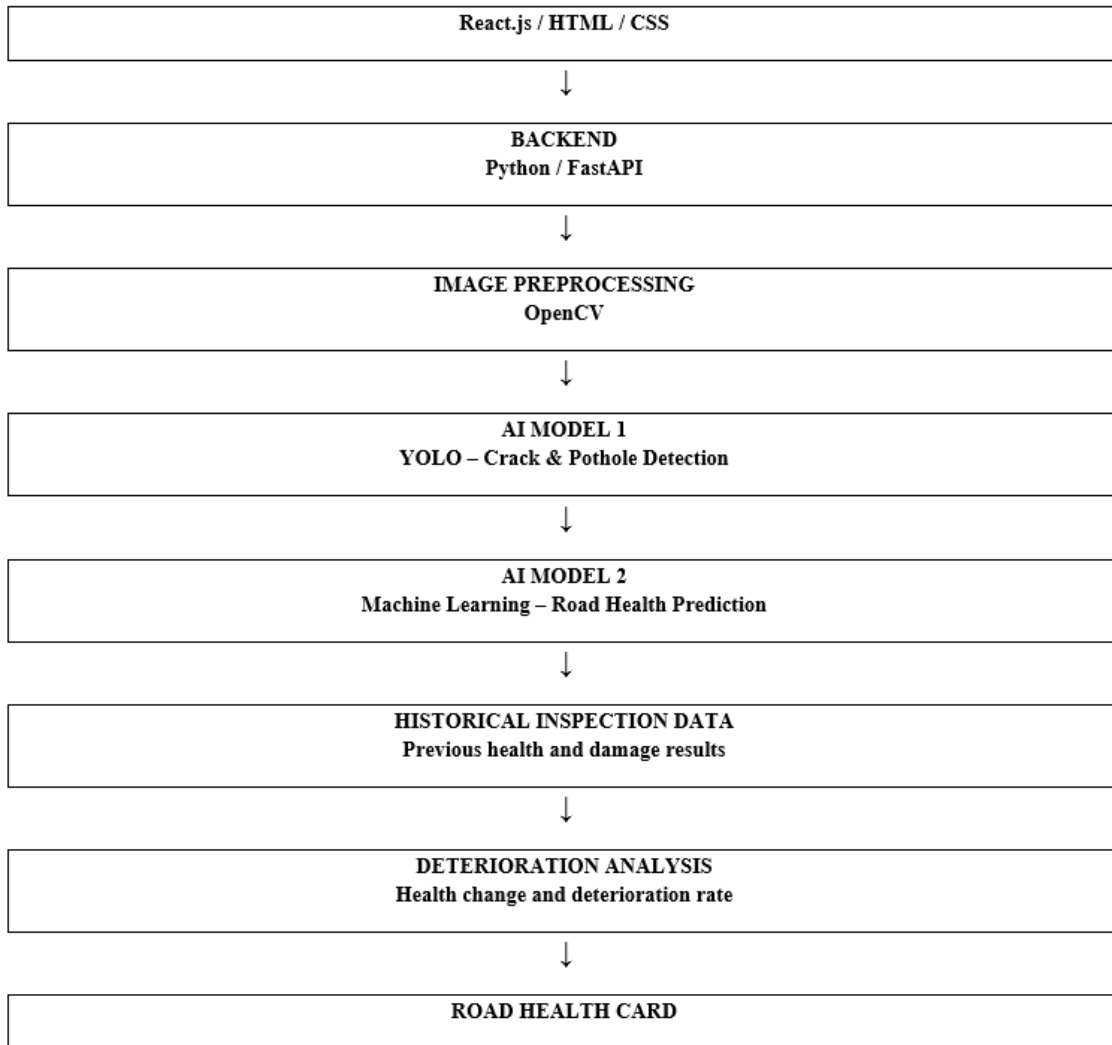
2.1 Project Description



3. System Architecture

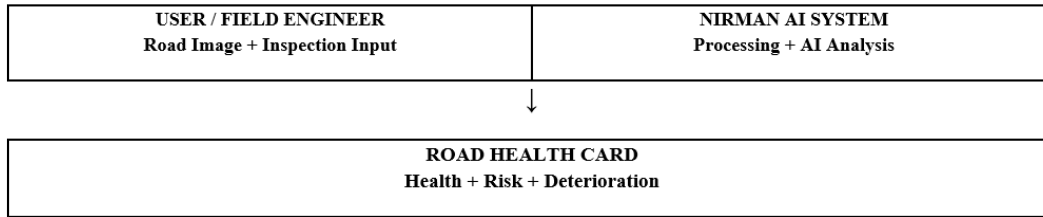
The system uses a modular architecture consisting of image input, preprocessing, Computer Vision, health prediction, historical inspection storage and deterioration analysis.

3.1 System Architecture – Block Diagram

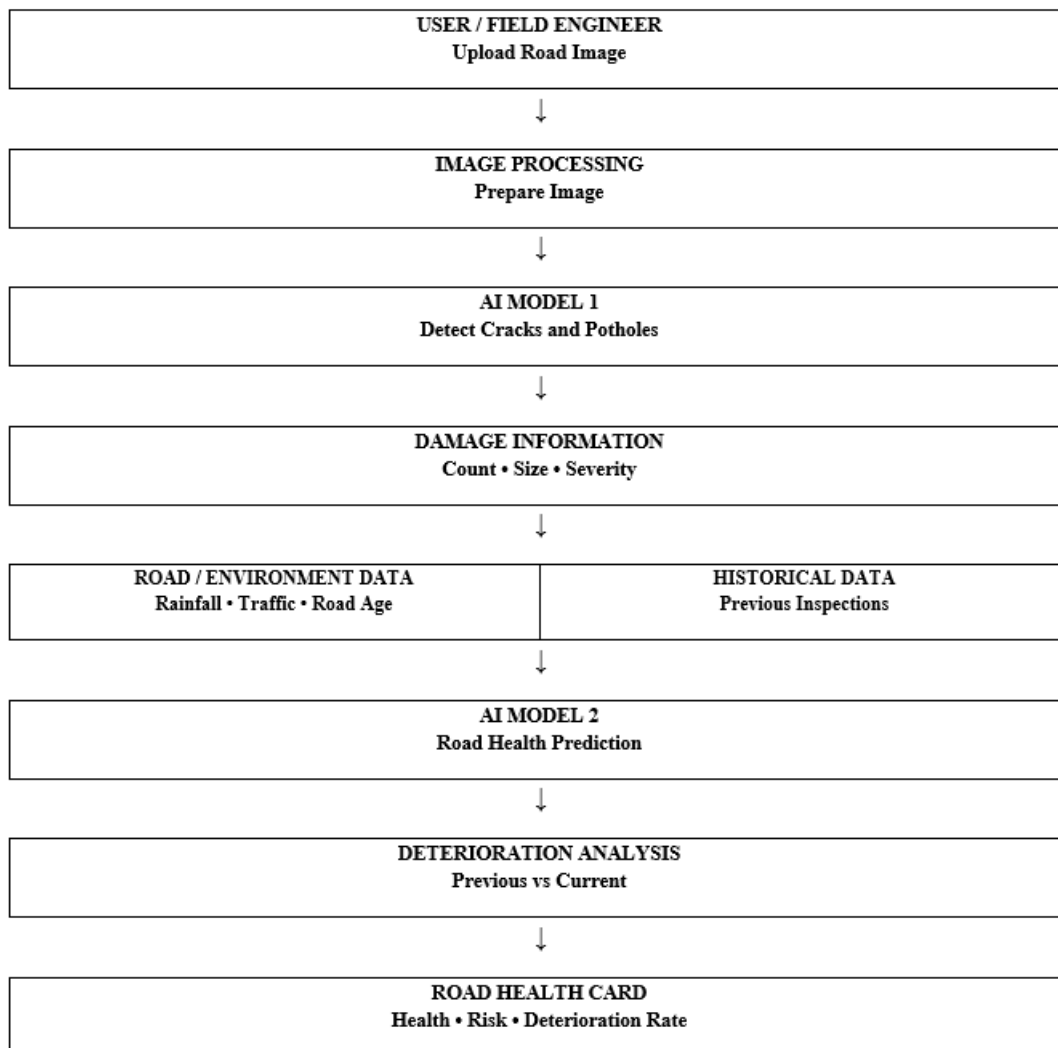


4. Data Flow Diagram (DFD)

1.1 4.1 Level 0 DFD – Context Level



1.2 4.2 Level 1 DFD



5. Features Implemented / Planned

- Road image upload
- Crack detection
- Pothole detection
- Crack and pothole counting
- Damage size/severity estimation
- 50-meter road segment analysis
- Road health score
- Historical inspection storage
- Health score comparison
- Deterioration rate and trend
- Road Health Card generation
- Future vehicle-mounted automatic image collection
- Future government, GIS, satellite, drone and IoT integration

6. Challenges Faced

- Lighting and camera angle can affect detection.
- Thin cracks can be difficult to identify.
- Real-world size measurement from a normal image is challenging.
- Traffic and rainfall data may not be available for every segment.
- The same 50-meter segment must be identified consistently across inspections.
- Reliable deterioration analysis requires repeated historical observations.
- AI results require real-world validation before engineering use.
- Large-scale automatic image collection will require suitable cameras and processing infrastructure.

7. Conclusion

NIRMAN AI provides an AI-based approach for road damage assessment and deterioration monitoring. Computer Vision identifies cracks and potholes from road photographs, while available road information can support current road-condition estimation.

The main additional feature is tracking the same 50-meter road segment across repeated inspections. By comparing historical and current health scores, the system estimates deterioration rate and highlights segments that are getting worse rapidly.

The current photo-upload prototype can be developed using open datasets and free resources. In future, vehicle-mounted cameras can automate data collection, while government data, GIS, satellite imagery, drones and IoT sensors can expand the system into a larger infrastructure monitoring platform.

8. Future Scope

- Automatic road-image collection through government inspection vehicles
- Satellite and GIS integration
- Drone-based inspection
- Authorized NHAI/PWD/government data integration
- IoT-based infrastructure sensors
- Long-term deterioration prediction using larger historical datasets
- Government dashboard for inspection and maintenance prioritization

